

INTRODUCTION TO DATA SCIENCE

Data Fundamentals

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Universidad Distrital Francisco José de Caldas

2026-I



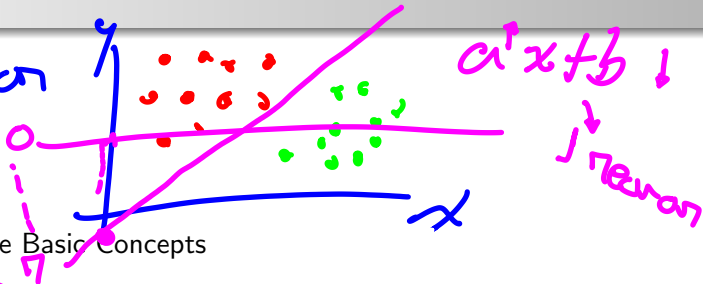
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- 1 Data Science Basic Concepts
- 2 What is to be a Data Scientist



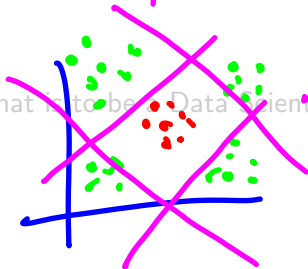
Outline

regression

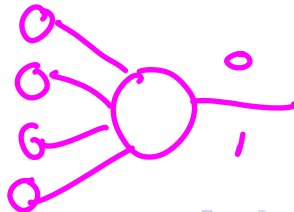


1 Data Science Basic Concepts

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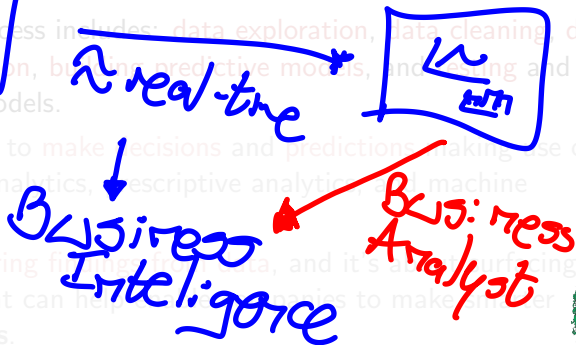
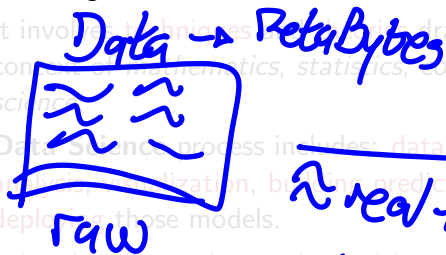


4 neurons



What is Data Science?

- Data Science** is an **interdisciplinary** field that uses **scientific methods, processes, algorithms**, and systems to **extract knowledge** and insights from structured and unstructured data.



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- It involves **techniques** and **theories** drawn from many fields within the context of **mathematics, statistics, computer science**, and **information science**.

Clustering → Gravity

• Data Science process includes: data exploration, data cleaning, data analysis, model building, building predictive models, and testing and deploying those models.

- It is primarily used to **make decisions** and **predictions** making use of predictive causal analytics, prescriptive analytics, and machine learning.
- It is about **uncovering findings from data**, and it's about surfacing hidden insights that can help enable companies to make smarter **business decisions**.



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- **Data Science** process includes **data exploration**, **data cleaning**, **data analysis**, **visualization**, **building predictive models**, and **testing and deploying** those models.

M.L.

Steps
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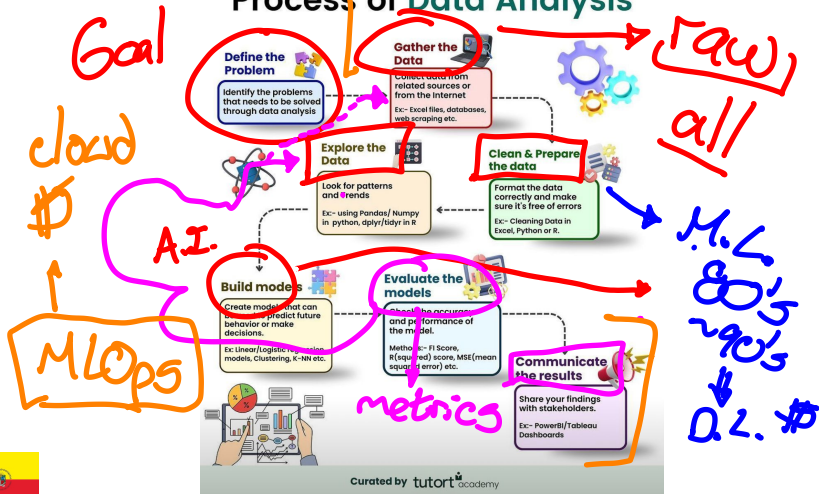
Patterns



Process of Data Analysis

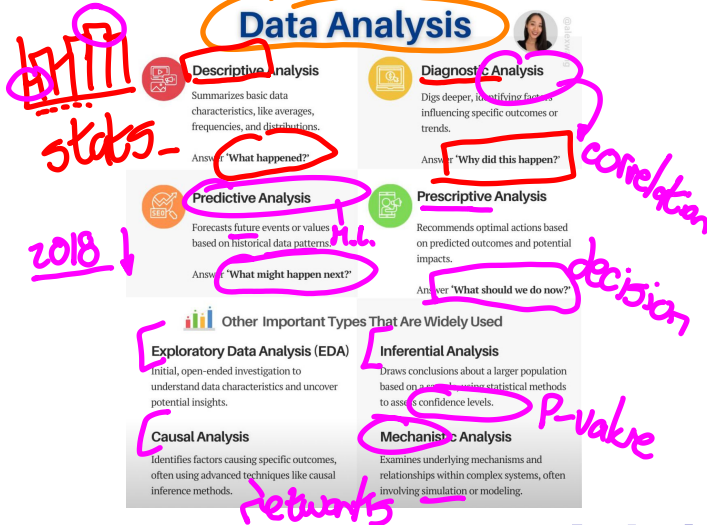

[Save for later](#)

Process of Data Analysis

Curated by tutort^u.academy

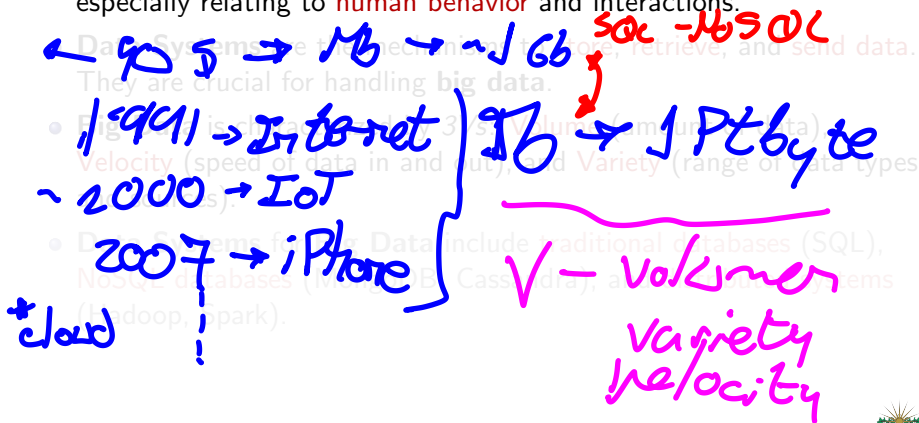
Types of Data Analysis

Different Types of Data Analysis



Data Systems & Big Data

- **Big Data** refers to extremely large data sets that may be analyzed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.



Data Systems & Big Data

Kaplan, Pabst, & Redman

- **Big Data** refers to **extremely large data sets** that may be analyzed computationally to reveal **patterns, trends**, and associations, especially relating to **human behavior** and interactions. **Stream**
- **Data Systems** are the mechanisms to **store, retrieve, and send data**. They are crucial for handling **big data**. **Data Lake**



Business

Data Mart

Hive
Presto
(Hadoop)
Spark

Parquet
(Apache)

relational

Data Warehouse
↓
Processed

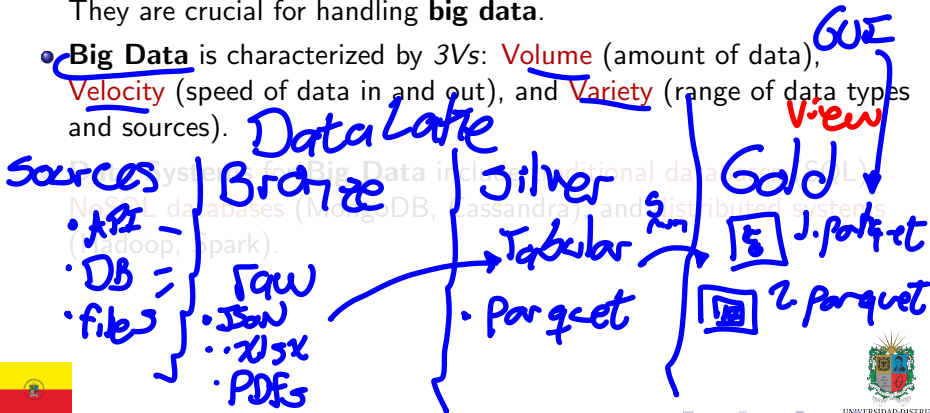


Data Systems & Big Data

store procedure



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- **Data Systems** for **Big Data** include **traditional databases (SQL)**, **NoSQL databases** (MongoDB, Cassandra), and **distributed systems** (Hadoop, Spark).

slow flake

Facebook

PySpark
↳ Pandas



History of Data

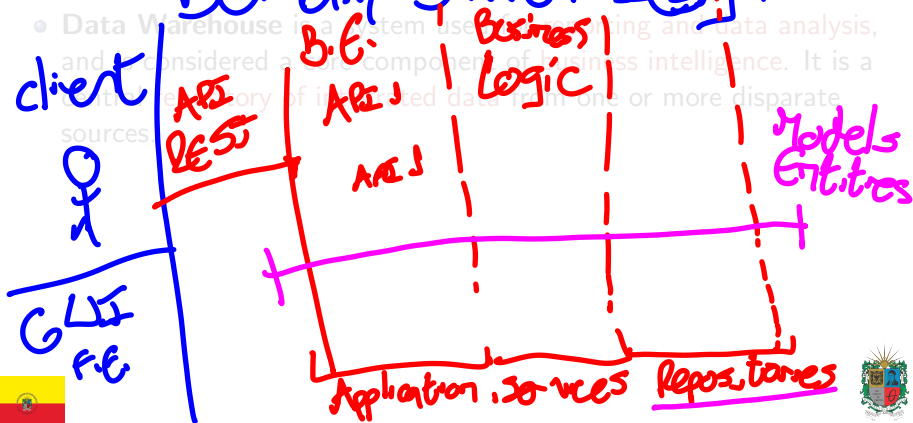


Data Lake Vs. Data Warehouse

cruda

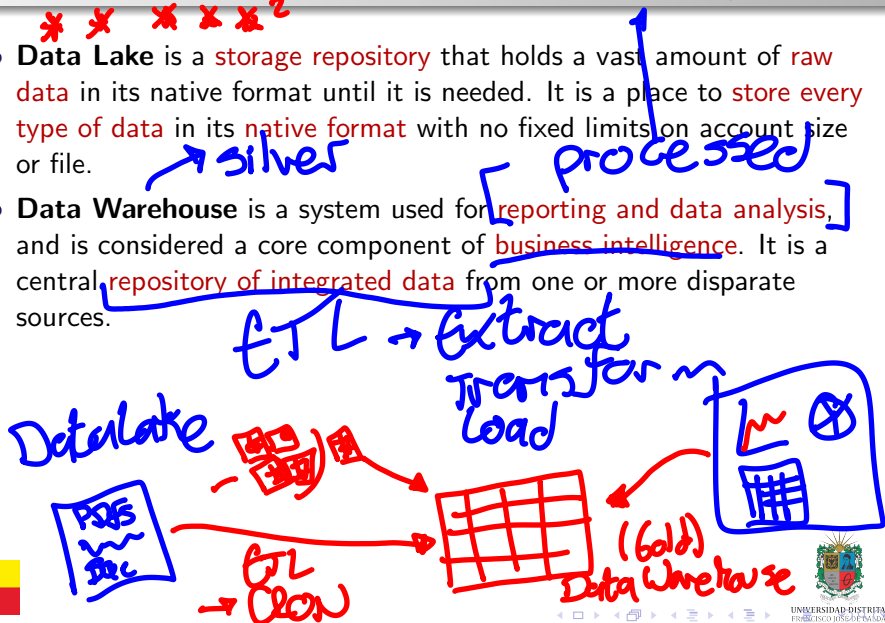
- **Data Lake** is a storage repository that holds a vast amount of raw data in its native format until it is needed. It is a place to store every type of data in its native format with no fixed limits on account size or file.

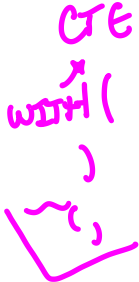
Domain Driven Design



Data Lake Vs. Data Warehouse

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- Data Warehouse** is a system used for **reporting and data analysis**, and is considered a core component of **business intelligence**. It is a central **repository of integrated data** from one or more disparate sources.





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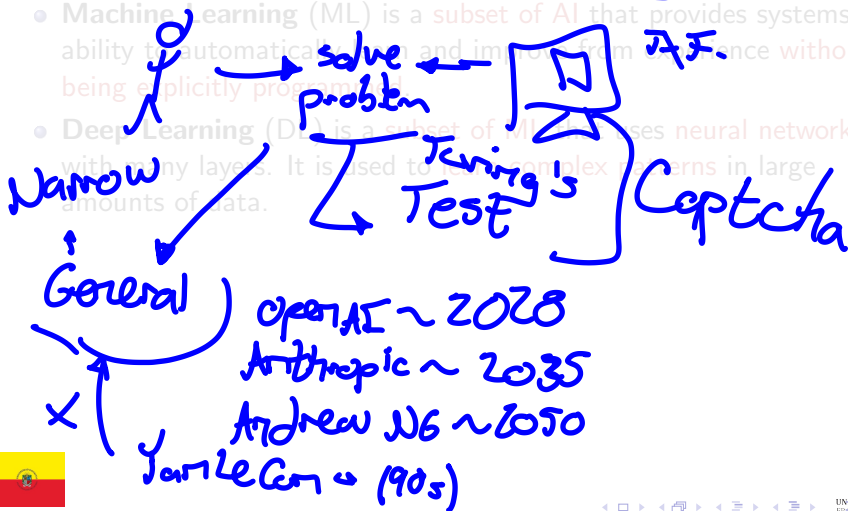
BitMap 

Mongo
 key value
 (JSON)
 $\sum \frac{1}{r} \cdot I(r, \theta)$
 love (r_2, θ_2)
 $L_2(r_3, \theta_3)$
 I love
 L1
 GPU
 θ



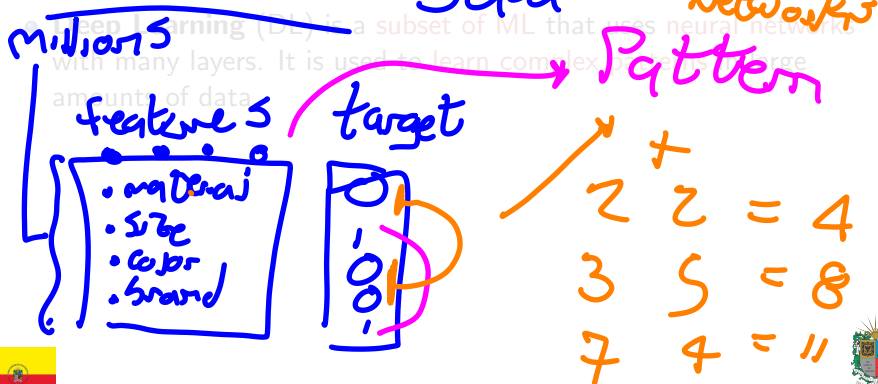
Artificial Intelligence, Machine Learning, Deep Learning

- **Artificial Intelligence (AI)** is the simulation of human intelligence processes by machines, especially computer systems.
- **Machine Learning (ML)** is a subset of AI that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.
- **Deep Learning (DL)** is a subset of ML that uses neural networks with many layers. It is used to learn complex patterns in large amounts of data.



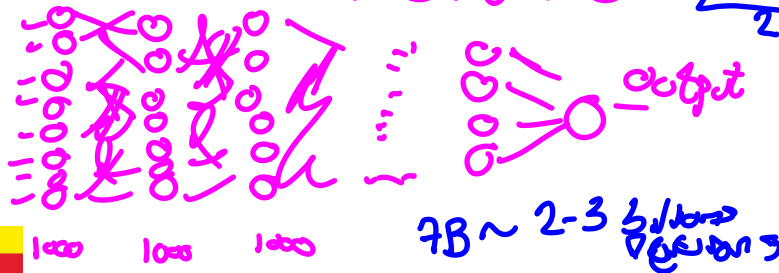
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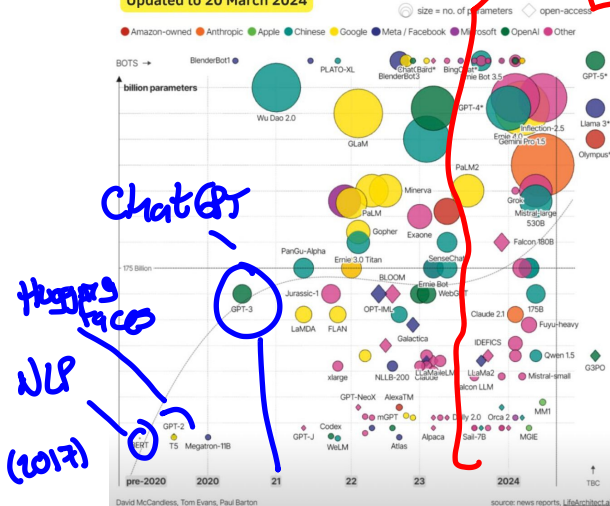


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Updated to 20 March 2024



Trillions

---! The stack
↓
(Curated
data



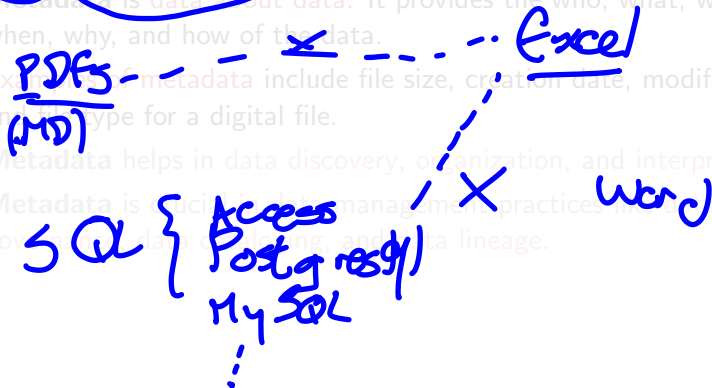
Data and MetaData

- **Data** refers to raw unprocessed, and unorganized facts or details that alone might not make much sense or provide context.

- Metadata is data about data. It provides the who, what, where, when, why, and how of the data.
- Examples of metadata include file size, creation date, modified date, and datatype for a digital file.

- Metadata helps in data discovery, organization, and interpretation.

- Metadata is crucial in data management practices in the government, data lineage, and data lineage.



Data and MetaData

raw $\xrightarrow{\text{lineage}}$ processed

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Data Governance $\rightarrow$ RACI Matrix

- Metadata helps in data discovery, organization, and interpretation.
- Metadata is critical for data management practices like data governance, data cataloging, and data lineage.

$\rightarrow$ Data Lineage (EL)

$\rightarrow$ Data Expectation (alerts)

- Data Owner
- Data Steward



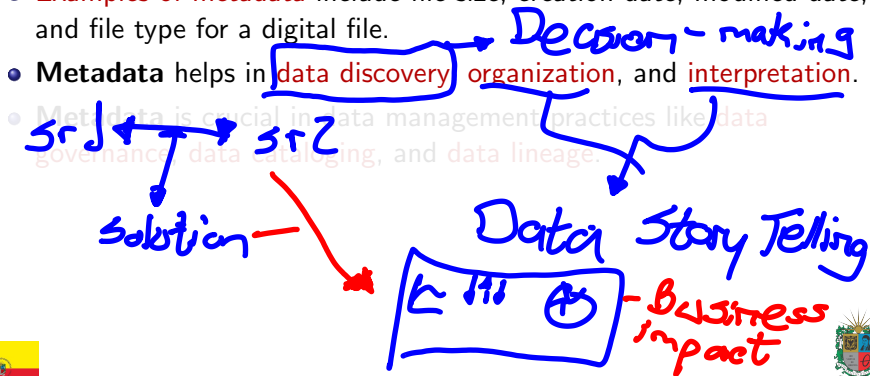
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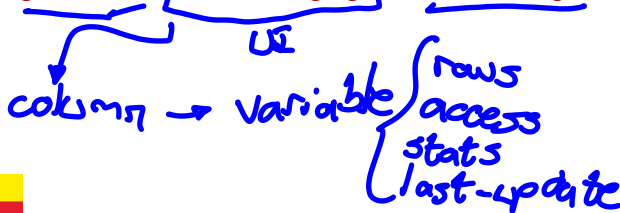
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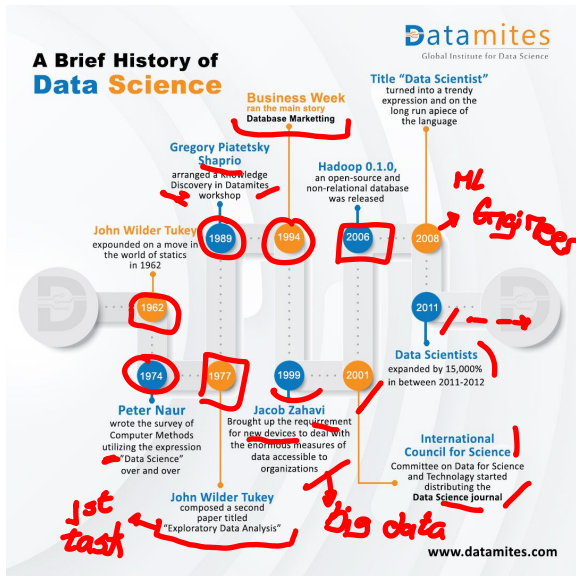


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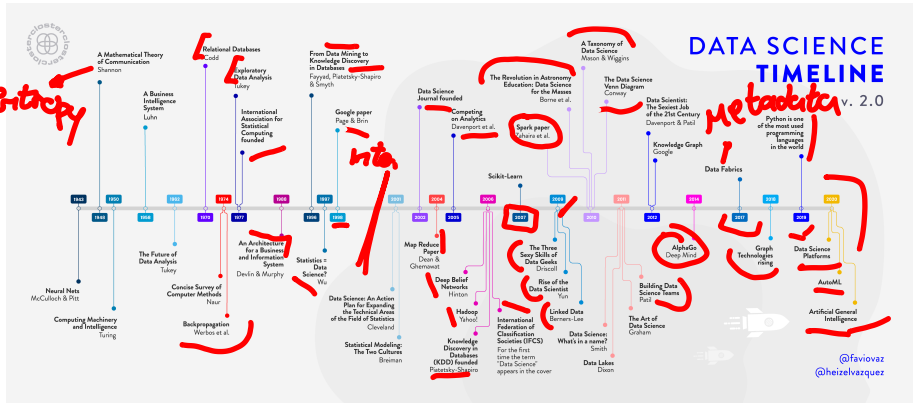
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Brief History of Data Science



Data Science Big Timeline



Data Science in Industry

- **Data Science** is used in **many industries** to **make decisions**, **optimize processes**, and **increase efficiency**.
- **Data Science** is used in healthcare to predict patient outcomes, optimize treatment plans, and personalize medicine.
- **Data Science** is used in finance to detect fraud, predict stock prices, and automate trading.
- **Data Science** is used in retail to optimize pricing, forecast demand, and personalize marketing.
- **Data Science** is used in manufacturing to predict equipment failures, optimize supply chains, and improve quality control.
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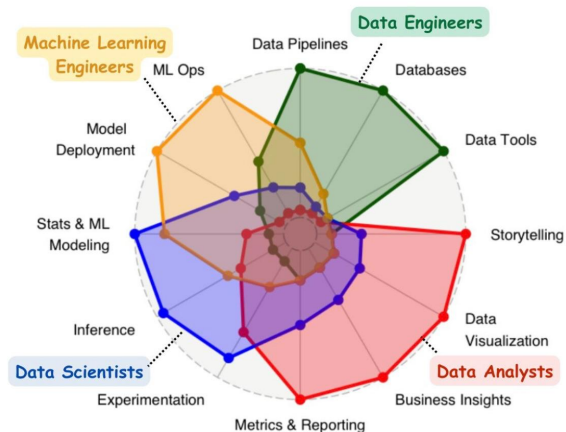
Outline

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Tech Team — Roles

Types of Data Roles - Where are you?



Tech Team — Data Profiles

WHICH PROFILE DESCRIBES YOU THE MOST?

ML ENGINEER



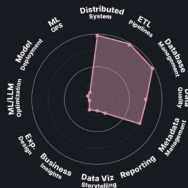
DATA STEWARD



DATA SCIENTIST



DATA ENGINEER



ANALYTICS ENGINEER



DATA ANALYST



Data Scientist Responsibilities

- Collecting large sets of structured and unstructured data from **disparate sources**.
- Cleaning and validating the data to ensure **accuracy**, **completeness**, and uniformity.
- Analyzing the data to identify **patterns** and trends.
- Interpreting the data to discover solutions and **opportunities**.
- Communicating findings to stakeholders using **visualization** and other means.
- Developing, prototyping, and implementing **machine learning models**.
- Staying current on techniques and tools in the field and continually **improving skills**.



Artificial Intelligence Tech Ecosystem

AI Infrastructure Tools open source

AI FRAMEWORKS, TOOLS & LIBRARIES



AI MODELS



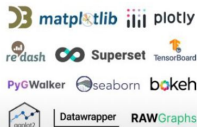
SEARCH



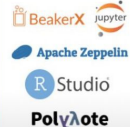
LOGGING & MONITORING



VISUALIZATION



COLLABORATION



@alexwang



Data Science Python Tech Ecosystem

Life is Short, I Use Python

Data Manipulation



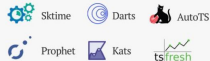
Statistical Analysis



Natural Language Processing



Time Series Analysis



Data Visualization



Machine Learning



Database Operations



Web Scraping



Data Sources and Formats

Data Definition Framework

Data Format

Data Source

Internal



Structured



Human-Generated

- Survey ratings
- Aptitude testing

Machine-Generated

- Web metrics from Web logs
- Product purchase from sales Records
- Process control measures

Unstructured



Human-Generated

- Emails, letters, text messages
- Audio transcripts
- Customer comments
- Voicemails
- Corporate video/communications
- Pictures, illustrations
- Employee reviews

External



Human-Generated

- Number of Retweets, Facebook likes, Google Plus +1s
- Ratings on Yelp
- Patient ratings ratings

Machine-Generated

- GPS for tweets
- Time of tweet/updates/postings

Human-Generated

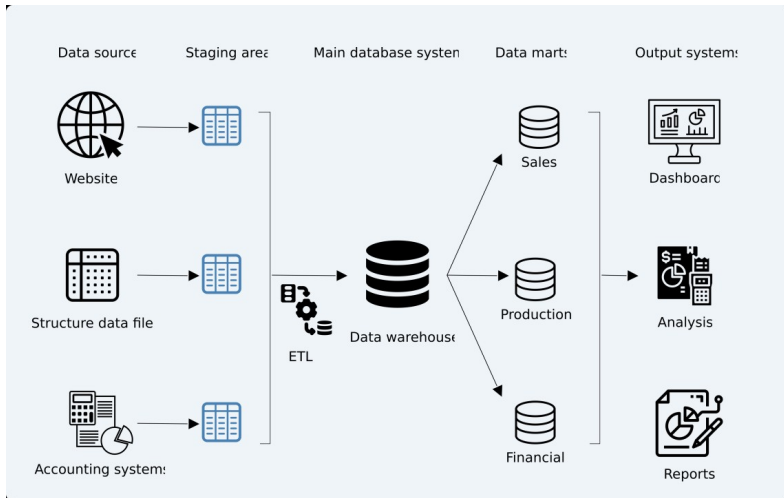
- Content of social media updates
- Comments in online forums
- Comments on Yelp
- Video reviews
- Pinterest images
- Surveillance video



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Data Pipelines



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

